

No. of Printed Pages : 4
Roll No.

221764C

6th Sem/ Mechanical Engineering
Sub. : Power Plant Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The area under the load curve represents ____.(CO1)
- a) The average load on power system
 - b) Maximum demand
 - c) Number of units generated
 - d) Load factor
- Q.2 The most commonly used moderator in nuclear plants is (CO4)
- a) Heavy water b) Concrete and bricks
 - c) Concrete d) Graphite
- Q.3 What is a major component of a gas turbine? (CO4)
- a) Boiler b) Condenser
 - c) Compressor d) Reactor
- Q.4 Which of the following is a type of air pollution from thermal power plants? (CO6)
- a) Acid rain b) Smog
 - c) Oxides of nitrogen d) All of the above

- Q.5 What is a method of nuclear waste disposal? (CO4)
- a) Electrostatic precipitation
 - b) High level waste disposal
 - c) Chain grate stoker
 - d) Ball mill
- Q.6 What is a disadvantages of thermal power plant (CO3)
- a) It uses nuclear fuel
 - b) It causes water pollution
 - c) It uses wind energy
 - d) It uses solar enrgy

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define nuclear fusion and nuclear fission? (CO4)
- Q.8 Write full form of "HAWT". (CO5)
- Q.9 Define "Base Load" in the power plant. (CO1)
- Q.10 ____ used as fuel in diesel power plant. (CO5)
- Q.11 Give one example of a pollutant from thermal power plants. (CO6)
- Q.12 Name the types of reactors used in nuclear power plant. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define (CO1)
- | | |
|--------------|-----------------|
| a) Peak load | b) Load Factor. |
| c) MPPT | d) HAWT |
- Q.14 Write the advantages and disadvantages of nuclear power plant. (CO4)
- Q.15 Explain pulverized fuel handling system. (CO3)
- Q.16 Write short note on solar Power Plant. (CO4)
- Q.17 Write the advantages of diesel power plant. (CO5)
- Q.18 Discuss the effect of global warming and its control. (CO6)
- Q.19 Briefly describe the methods for nuclear waste disposal. (CO4)
- Q.20 Differentiate between impulse and reaction turbine. (CO2)
- Q.21 Describe the general layout of thermal power plant in brief. (CO3)
- Q.22 Write the advantages of hydro-electric power plant. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the working of open and closed cycle gas turbine with neat diagram. (CO4)
- Q.24 Explain the coal handling system in a thermal power plant, including storage and fuel burning methods. (CO3)
- Q.25 Discuss the environmental impact of thermal power plants, including air, water and noise pollution and methods to control them. (CO6)